

Skills Summary

Modeling with Systems and Matrices

Use this reference while completing your worksheet or when studying.

Skill 1 — Systems in two unknowns

Choose the method from the shape of the equations. If one equation is already solved for a variable (or costs one step to solve), substitute. If both are in the $ax + by = c$ form with awkward coefficients, eliminate by scaling one equation until a column matches.

Always define the variables in words first — “ x = price of one adult ticket, in dollars”. A system you cannot label is a system you cannot check.

Example: Solve $2x + y = 13$ and $x - y = 2$.

Step 1. The y terms already have opposite signs, so adding the two equations removes y in one move — elimination is cheaper here than substitution.

Step 2. Adding: $3x = 15$, so $x = 5$. Then $5 - y = 2$.

$$\Rightarrow x = 5, y = 3$$

Try it: Solve $3x + 2y = 16$ and $x + y = 6$.

check: $x = 5, y = 3$

Skill 2 — Systems in three unknowns

Reduce three to two. Pair the equations up and eliminate the *same* variable from each pair. That leaves a two-unknown system you already know how to finish; back substitute for the third.

Look for a free gift first: an equation like $y = x + 3$, or two equations that differ in only one term, does the elimination for you.

Example: Solve $x + y + z = 9$, $2x + y + z = 11$, $x + 2y + z = 13$.

Step 1. The three equations differ from the first in a single term each, so subtracting the first from the others isolates one unknown at a time.

Step 2. Second minus first: $x = 2$. Third minus first: $y = 4$.

Step 3. Back into the first: $2 + 4 + z = 9$.

$$\Rightarrow x = 2, y = 4, z = 3$$

Try it: Solve $x + y + z = 12$, $x + 2y + z = 15$, $2x + y + z = 16$.

check: $x = 2, y = 4, z = 3$

Skill 3 — Writing a system as a matrix equation

The translation is mechanical: coefficients fill the matrix, one row per condition and one column per unknown; the unknowns stack into a column; the right-hand sides stack into another. That is $A\mathbf{x} = \mathbf{b}$.

What it buys you is a picture of the model: each row is one real-world condition, each column is one real-world quantity. Solve it with the same elimination as before.

Example: Write $4x + 3y = 18$ and $2x + 5y = 16$ as a matrix equation, and solve.

Step 1. Each condition becomes one row, so the unknowns line up in columns and the constants stand alone on the right.

Step 2.
$$\begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 18 \\ 16 \end{bmatrix}$$

Step 3. Doubling the second row gives $4x + 10y = 32$; subtracting the first row leaves $7y = 14$.

$$\Rightarrow x = 3, y = 2$$

Try it: Write $5x + 2y = 24$ and $3x + 4y = 20$ as a matrix equation, and solve.

check: $\mathbf{z} = \mathbf{A}\mathbf{x} = \mathbf{b}$

Skill 4 — Where a line meets a curve

Substitute the linear equation into the nonlinear one. One quadratic in one unknown comes out, so expect *two* answers — and each x carries its own y .

Finish honestly: pair the coordinates up, then ask whether the situation allows both. A negative length is rejected by the context; a second valid design is not.

Example: Where does $y = 2x + 4$ meet $y = x^2 + 1$?

Step 1. Both equations give y , so setting the right-hand sides equal turns the pair into one quadratic — the standard move for a line against a curve.

Step 2. $x^2 + 1 = 2x + 4 \Rightarrow x^2 - 2x - 3 = 0 \Rightarrow (x - 3)(x + 1) = 0$

Step 3. At $x = 3$: $y = 2(3) + 4 = 10$. At $x = -1$: $y = 2(-1) + 4 = 2$.

$$\Rightarrow x = 3, y = 10 \text{ and } x = -1, y = 2$$

Try it: Where does $y = 3x$ meet $y = x^2 - 4$?

check: $\mathbf{y} = \mathbf{A}\mathbf{x} = \mathbf{b}$ and $\mathbf{z} = \mathbf{A}\mathbf{x} = \mathbf{b}$